

# Jeril Kuriakose

Senior Principal Engineer — Generative AI / LLMs



Deep expertise in LLMs, NLP and agentic AI systems. I lead end-to-end development of high-impact platforms — from data engineering and training through MLOps, inference optimisation and secure deployment at scale.

Currently at HUMAIN as Senior Principal Engineer, AI Research, leading the inference platform for ALLaM and open-weight LLMs and the Arabic voice (ASR/TTS) platform. Open to principal and staff-level roles in applied AI.

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14

YEARS IN DATA & AI

100K

PERSON-HOURS SAVED / YR

35+

PEER-REVIEWED PAPERS ➦

## 01 Practice

Current focus and working stack

### SUMMARY

Currently at HUMAIN, building the production inference platform for **ALLaM** and open-weight LLMs and the Arabic voice (ASR/TTS) platform: multi-node serving on Kubernetes with vLLM and SGLang, model-lifecycle automation, and MLOps for latency and cost.

Earlier, owned the **ALLaM** post-training stack at SDAIA: curriculum SFT, iterative DPO, and RL-SFT hybrid training. Proven record of measurable outcomes and hands-on leadership of cross-functional teams.

### HIGHLIGHTS

1M+

USERS REACHED

+61%

LLM SERVING THROUGHPUT

16

MODELS IN PRODUCTION

0.18 WER

0.03 RTF  
REAL-TIME ASR

### ML / AI

LLMs   NLP   Speech (ASR/TTS)   Agentic systems   RAG   Time-series

### FRAMEWORKS & SERVING

Python   PyTorch   Transformers   TRL   Data-Juicer   vLLM / SGLang / Triton   Ray  
LangGraph / Dify

### MLOPS, INFRA & PLATFORM

Kubernetes   GitOps (FluxCD / Kustomize)   Kong   Kafka   Metabase   Azure / GCP  
MLflow   ClickHouse / Qdrant

### AI-AUGMENTED DEV

Claude Code   OpenCode   Codex   Custom skills & agents

**Senior Principal Engineer, AI Research** HUMAIN CURRENT

Riyadh, Saudi Arabia Apr 2025 – Present

Owens technical strategy and hands-on delivery for HUMAIN's LLM inference and Arabic voice platforms: distributed serving, model-lifecycle automation, evaluation, production readiness, and latency and cost optimisation.

- Architected and operate HUMAIN's GitOps-managed **multi-model inference platform** on Kubernetes (H100), with standardised deployment recipes and a secured API gateway (JWT/ACL), plus a self-hosted **ALLaM** playground on OpenWebUI, serving ALLaM and open-weight models to internal and external consumers.
- Established **multi-node inference** for MoE models up to 744B parameters, using vLLM/Ray and SGLang/NCCL execution patterns with tensor and pipeline parallelism across multi-node H100 clusters.
- Built an experiment-driven **serving-optimisation** practice: a serving-config change improved TTS time-to-first-audio 6.5x and throughput **38%**, speculative decoding delivered up to 3.1x throughput at low concurrency, and designed autoscaling and workload tiering to reduce idle GPU capacity.
- Built the **deployment and validation tooling** behind the platform: a benchmarking harness (TTFT and concurrency sweeps, GPU sizing), a validated per-model recipe library, event-driven model onboarding, and automated rollout with post-deploy health gates.
- Led production engineering for HUMAIN's **Arabic voice** services on Triton: 8kHz telephony ASR (wav2vec2; WER, CER, real-time factor; diarisation and VAD) and Arabic TTS (diacritisation, voice controls, output guardrails).
- Established the **voice evaluation and release discipline**: an internal voice-evaluation portal with rating and comparison workflows, TTS regression testing, and versioned ASR model, alias, and language-model mapping.
- Drove production rollout of the voice platform: API gateway with TLS, Keycloak IAM, Kafka eventing, metering and usage analytics, observability, ISO/GRC control alignment, and a cross-cloud migration.
- Produced executive product analytics: **Metabase** dashboards and a prompt-label study over 10M+ messages, covering MAU/DAU, retention cohorts, engagement and loyalty segments, work-versus-leisure use, intent, and 24 capability categories.

**Principal Data Scientist** Saudi Data & AI Authority

SDAIA · Riyadh, Saudi Arabia Jan 2024 – Apr 2025

Owens the **post-training and inference stack for ALLaM**, Saudi Arabia's national Arabic large language model, from **50TB**-scale data processing through 70B-parameter production serving.

#### Saudi Data & AI Authority (SDAIA) (continued)

- Lead data processing and inference optimisation for **ALLaM** (Arabic LLM); manage training and inference stacks across PyTorch, Transformers, TRL, vLLM/TGI/Triton, Ray, Kubernetes and Azure.
- Built the post-training stack for agentic capability: interleaved RL with periodic SFT injections, iterative DPO with uncertainty-aware preference collection, self-optimised fine-tuning via self-distillation, and gradient separation across RL/SFT objectives.
- Designed a curriculum SFT pipeline for agentic behaviour; improved model convergence by **~35%**.
- Processed **50TB** of RedPajama-Data-v2 through Data-Juicer; implemented LLaMA-2 pretraining strategies, both from scratch and continual.
- Deployed and optimised ALLaM **7B / 13B / 70B** on Kubernetes; identified optimal inference parameters per use case.
- Implemented agentic RAG and multi-agent planning and orchestration with LangChain, LangGraph and Dify.
- Designed AI-augmented development workflows on Claude Code, OpenCode and Codex; authored custom skills and agents that automate engineering tasks and shorten delivery.

#### Senior Data Scientist **Mizuho Bank**

Singapore Jul 2019 – Dec 2023

Built and productionised **four named NLP and forecasting platforms** for a global investment bank, and introduced the organisation's first model-tracking practice.

- **Phoenix** — generic NLP information-extraction platform on BERT with a hybrid CNN and Bloom embeddings; **~90%** accuracy; saved **~100,000 person-hours per year**.
- **PIGEON** — document classification with RoBERTa plus summarisation with GPT-2 and GPT-Neo; **97%** accuracy; full API productionisation.
- **SWAN** — anti-money-laundering name screening; cut false positives by **20%**; vector database for embeddings and graph neural networks for anomaly detection.
- **HAWK** — hybrid encoder-decoder LSTM for daily, weekly and monthly forecasts; saved **~3,000 person-hours per year**; external data ingestion with ELK.
- Introduced **MLflow** for organisation-wide model tracking; implemented ML monitoring and production performance validation.

#### EARLIER EXPERIENCE

|           |                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------|
| 2019      | <b>Senior Data Scientist</b> · Baker Hughes (a GE Company), Kochi                                       |
| 2018–2019 | <b>Senior AI Engineer</b> · Innovation Incubator, Thiruvananthapuram                                    |
| 2017–2018 | <b>Senior ML/AI Developer</b> · Raw Data Technologies, Kochi                                            |
| 2015–2017 | <b>Assistant Professor &amp; Data Analyst</b> · St. John College of Engineering and Technology, Palghar |

#### EARLIER EXPERIENCE (CONTINUED)

|           |                                                                        |
|-----------|------------------------------------------------------------------------|
| 2013–2015 | PhD Scholar, Developer & Data Analyst · Manipal University Jaipur      |
| 2012–2013 | Developer & Assistant Professor · Kavery College of Engineering, Salem |

## 03 AI-Augmented Engineering

Governed autonomous delivery and operations

- **Built a governed autonomous delivery system.** Agents take a ticket through triage, evidence-based planning, isolated implementation, tests, adversarial code review, and security review, then open a merge request for human approval. No auto-merge.
- **Enforce machine-checked readiness before any agent runs.** Code-aware certification pins each plan to the live codebase and a policy register, hash-bound, so stale or ambiguous work is blocked.
- **Operate a multi-runtime, multi-model harness** across Claude Code, Codex, Gemini, and OpenCode, with reusable agent skills, first-pass-rate telemetry, and phone-based control over Telegram.
- **Run unattended daily data and model pipelines** with validation, checkpointing, safe retries, and scheduled GPU scaling.

## 04 Selected publications

From 35+ total · see Google Scholar

- |      |                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>ALLaM: Large Language Models for Arabic and English</b><br>Bari M. S., Kuriakose J., et al. ICLR                                                                                           |
| 2023 | <b>A review of deep learning-based approaches for detection and diagnosis of diverse classes of drugs</b><br>Kuriakose J., et al. Archives of Computational Methods in Engineering (Springer) |
| 2022 | <b>EMBN-MANET: Eliminating Malicious Beacon Nodes in UWB-based Mobile Ad-Hoc Networks</b><br>Kuriakose J., Joshi S., Bairwa A. K. Ad Hoc Networks (Elsevier)                                  |
| 2015 | <b>Secure Multipoint Relay Node Selection in Mobile Ad Hoc Networks</b><br>Kuriakose J., Amruth V., Raju R. V. Communications in Computer and Information Science (Springer)                  |
| 2014 | <b>A Review on Mobile Sensor Localization</b><br>Kuriakose J., et al. Security in Computing and Communications (Springer)                                                                     |

EDUCATION

- Ph.D. in Computer Engineering**

Manipal University Jaipur, School of Computing and IT

Jul 2013 – Dec 2019
- M.Tech. in Information Technology**

ISIM, University of Mysore

Jul 2012
- B.Tech. in Information Technology**

Jeppiaar Engineering College (Anna University)

May 2010

LANGUAGES

|           |        |
|-----------|--------|
| Tamil     | NATIVE |
| Malayalam | NATIVE |
| English   | FLUENT |
| Hindi     | FLUENT |
| Arabic    | BASIC  |

RECOGNITION

-  **Project of the Year**

Mizuho Bank · Jul 2022
-  **Employee of the Month**

Mizuho Bank · Oct 2019, May 2021, Jul 2021
-  **Honourable Award – AML Project**

Mizuho Bank · Feb 2020
-  **Honourable Mention – Phoenix Project**

Mizuho Bank · Dec 2020